

Computer System Architecture

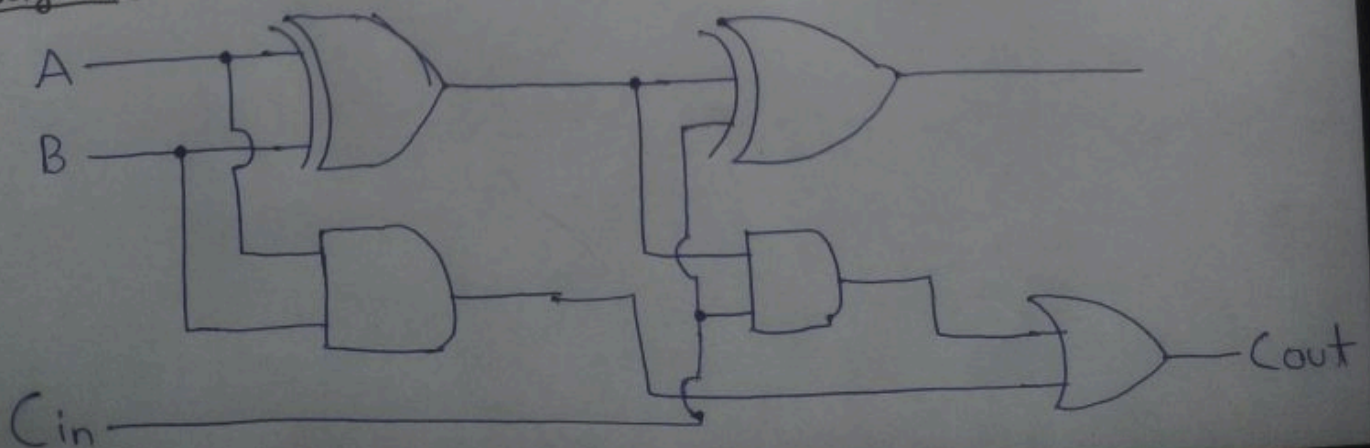
Assignment - 1

Q4. Implement a 2-bit adder using logic gates. Show the circuit diagram and truth table.

Ans. Truth Table of 2-bit adder:

A_1	A_0	B_1	B_0	S_1	S_0	Carry
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	1	0	0
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	1	0	0
0	1	1	1	1	1	0
1	0	0	0	0	1	0
1	0	0	1	1	0	0
1	0	1	0	1	0	0
1	0	1	1	1	1	0
1	1	0	0	1	0	0
1	1	0	1	1	1	0
1	1	1	0	1	1	0
1	1	1	1	1	1	0
1	1	1	1	1	1	1

★ Diagram:



Q2 Convert the given Standard POS to Canonical POS:

$$F(P, Q, R) = (P+Q) \cdot (Q+R) \cdot (P+R)$$

Ans. Missing Variable in each Term:

- $(P+Q)$ is missing R
- $(Q+R)$ is missing P
- $(P+R)$ is missing Q

$$\star X+YZ = (X+Y)(X+Z)$$

- $(P+Q) = (P+Q+R)(P+Q+R')$
- $(Q+R) = (P+Q+R)(P'+Q+R)$
- $(P+R) = (P+Q+R)(P+Q'+R)$

$$F(P, Q, R) = (P+Q+R)(P+Q+R')(P+Q+R)(P'+Q+R)(P+Q+R)(P+Q'+R)$$

$$\Rightarrow F(P, Q, R) = (P+Q+R)(P+Q+R')(P'+Q+R)(P+Q'+R)$$

Q3. Minimize using Karnaugh Map:

$$F(A, B, C, D) = \prod(0, 2, 5, 7, 9, 11)$$

Ans.

	CD			
AB	00	01	11	10
00	1	1	0	0
01	0	1	1	0
11	0	1	1	0
10	1	0	0	0

Q4 Perform the following:

a. $1010 + 1100$

b. $(11101.00110)_2 = (\quad)_8 = (\quad)_{10}$

$$\begin{array}{r} \text{a. } 1010 \\ + 1100 \\ \hline 10110 \end{array}$$

$$\bullet (1010)_2 + (1100)_2 = (10110)_2$$

b. Convert to Octal (Base 8)

$$11101_2 \rightarrow \underline{011} \ \underline{101}$$

$$\bullet 011_2 = 3_8$$

$$\bullet 101_2 = 5_8$$

$$11101_2 = 35_8$$

$$\star 00110_2 \rightarrow \underline{001} \ \underline{100}$$

$$\bullet 001_2 \rightarrow 1_8$$

$$\bullet 100_2 \rightarrow 4_8$$

$$00110_2 \rightarrow .14_8$$

$$(11101.00110)_2 = (35.14)_8$$

$\star$ Convert to Decimal (Base 10)

$$11101_2 = (1 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

$$29 = 16 + 8 + 4 + 0 + 1$$

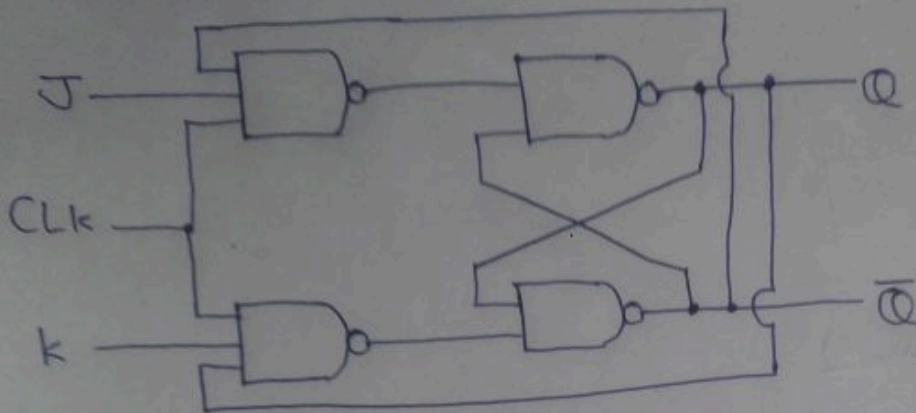
$$00110_2 = (0 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3}) + (1 \times 2^{-4}) + (0 \times 2^{-5})$$

$$= 0 + 0 + 0.125 + 0.0625 = 0.1875$$

$$(11101.00110)_2 = (29.1875)_{10}$$

Q 5. Draw the block diagram of JK Flip-flop along with its truth table, Characteristic table and excitation table?

Ans. Block Diagram:



★ Truth Table:

J	K	$Q(t)$	$Q(t+1)$	Description
0	0	Q	Q	No change (Hold)
0	1	Q	0	Reset
1	0	Q	1	Set
1	1	Q	Q'	Toggle

★ Characteristic Table

J	K	$Q(t)$	$Q(t+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

★ Excitation Table

$Q(t)$	$Q(t+1)$	J	K
0	0	0	0
0	1	1	0
1	0	0	1
1	1	1	1

Q6. What are the different types of Registers?

Ans. 1. Accumulator Register (AC)

- Stores intermediate results of arithmetic and logic operations
- Directly interacts with the Arithmetic Logic Unit (ALU)

2. Address Register (AR)

- Holds memory address of data or instructions to be fetched from or stored.
- Works closely with the Memory Data Register (MDR)

3. Data Register (DR)

- Temporarily holds data fetched from or written to memory
- Works as a buffer between CPU and Memory.

4. Program Counter (PC)

- Holds the address of the next instruction to be executed.
- Automatically increments after each instruction fetch.

5. Instruction Register (IR)

- Stores the currently executing instruction.
- The Control Unit (CU) decodes the instruction from this register.

6. Buffer Register

- Temporarily stores data being transferred between components.
- Used in I/O operations for smooth data flow.

7. Input Register

- Holds data received from an input device before processing

8. Output Register

- Holds data before sending it to an output device.

Q7. What is the purpose of a Program Counter (PC) register in a Computer System?

Ans. ★ Main Functions of the Program Counter (PC):

1. Holds the address of the Next Instruction
 - The PC Stores the memory address of the next instruction that needs to be fetched and executed by the CPU.
2. Ensures Sequential Execution of Instruction
 - After fetching an instruction, the PC automatically increments to point to the next instruction in memory.
 - This allows the program to execute instructions Sequentially.
3. Facilitates Jumps, Loops, and Branching
 - If a jump (JMP), call, or branch instruction is encountered, the PC is updated to the new address instead of incrementing Sequentially.
4. Works with the Control Unit (CU)
 - The CU fetches the instruction from memory using the address stored in the PC.
 - The instruction is then stored in the Instruction Register (IR) for decoding and execution.
5. Resets to the Starting Address on System Boot.
 - When a computer starts, the PC is initialized to point to the first instruction of the OS (often stored in ROM or BIOS)

Q8. Describe the different types of buses in a Computer System with the help of a diagram. Explain their Functions and importance.

Ans. Types of Buses:

- A Computer System has three main types of buses:

1. Address Bus

2. Data Bus

3. Control Bus

★ Address Bus:

- Definition: A unidirectional bus that carries the memory address of data or instructions.
- The CPU uses the address bus to specify the location in memory or I/O where data needs to be fetched or stored.
- The width of the address bus determines the maximum addressable memory.
- Unidirectional: Data flows only from CPU to memory or I/O devices.

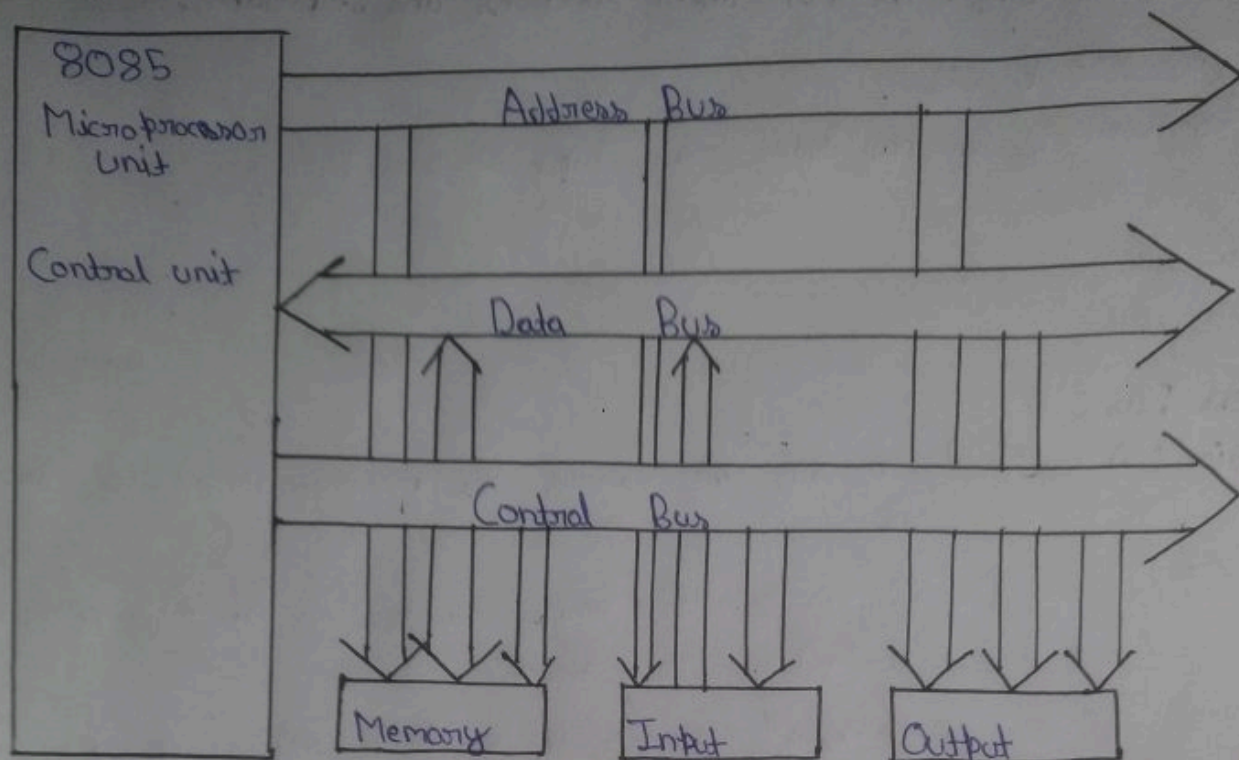
★ Data Bus:

- Definition: A bidirectional bus that transfers actual data between CPU, Memory, and I/O devices.
- The CPU reads data from memory or writes data to memory using the data bus.
- The width of the data bus determines how much data can be transferred at a time.
- Bidirectional: Data can move to and from the CPU.

★ Control Bus:

- Definition: A bidirectional bus that carries control signals between CPU and other components.
- Sends control signals like Read, Write, Interrupt, Clock, Reset, to manage operations.
- Ensures proper coordination between CPU, Memory, and I/O devices.
- Bidirectional: CPU sends and receives control signals.

★ Diagram of Computer Bus System



Q9. Give two differences between Combinational Circuit and Sequential Circuit.

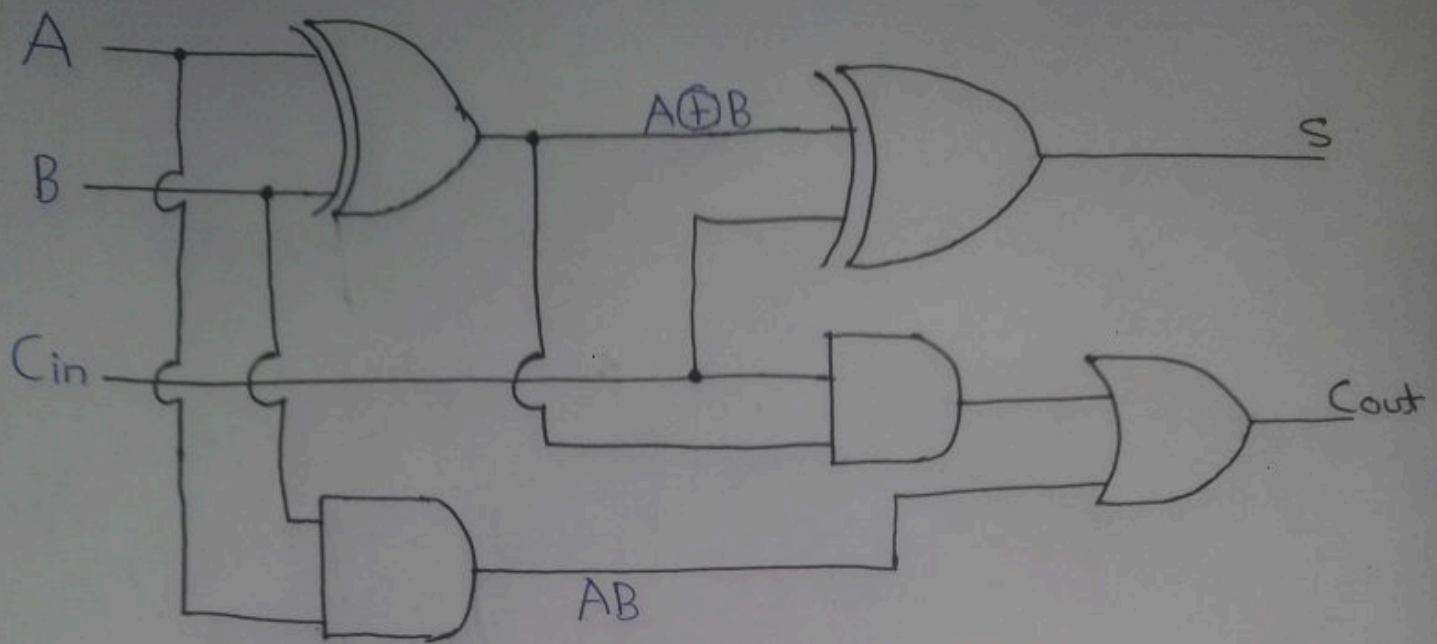
Ans Memory:

- **Combinational Circuit**: Does not have memory; the output depends only on the current inputs.
- **Sequential Circuit**: Has memory; the output depends on both the current inputs and past inputs.

★ Clock Dependency:

- **Combinational Circuit**: Does not require a clock signal.
- **Sequential Circuit**: Requires a clock signal to control state transitions.

Q10 Implement full adder using Half adders?



Logic Diagram of Full Adder using Half Adders